

Machine Learning Tutorial

PAC Learning, VC Dimension, GLMs, Correlated Regression, and Regularization

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1 The Basic Learning Problem

1.1 From Data to a Classifier

Consider a binary classification problem such as detecting whether an email is spam. Let $\mathcal{X}$ denote the input space. An input $x \in \mathcal{X}$ represents an email, possibly described by features such as the number of links, presence of certain words, sender information, etc.

The examples are generated according to an unknown probability distribution D over the input space. Thus, $x_i \sim D$. We assume that there exists an unknown target labeling function

$$f : \mathcal{X} \rightarrow \{0, 1\},$$

where, for example,

$$f(x) = \begin{cases} 1, & \text{if } x \text{ is spam,} \\ 0, & \text{if } x \text{ is not spam.} \end{cases}$$

The learner does not know f or D . Instead, it receives a finite training sample

$$S = \{(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)\},$$

where $x_i \stackrel{\text{i.i.d.}}{\sim} D$ and, in the realizable setting, $y_i = f(x_i)$. A learning algorithm A takes the training sample S as input and returns a hypothesis $h = A(S)$, where $h : \mathcal{X} \rightarrow \{0, 1\}$.

The fundamental question of statistical learning theory is therefore

How well will the learned hypothesis perform on unseen examples? The training data alone are not sufficient to answer this question. We need a connection between performance on the observed data and performance on new data.

This is the central motivation for **PAC learning** and **VC dimension**.

1.2 Training Error and True Error

There are two fundamentally different notions of error.

1.2.1 Training Error

The **training error**, also called the **empirical risk**, measures how well a hypothesis performs on the training sample.

For binary classification, using the zero-one loss, $\ell(h(x), y) = \mathbf{1}[h(x) \neq y]$, the empirical risk is $\hat{R}_S(h) = \frac{1}{m} \sum_{i=1}^m \mathbf{1}[h(x_i) \neq y_i]$.

For example, if a classifier makes mistakes on 5 out of 100 training examples, then

$$\hat{R}_S(h) = \frac{5}{100} = 0.05.$$

Thus, its training error is 5%.

1.2.2 True or Population Risk

The quantity we actually care about is the error on future, unseen examples.

The **true risk**, also called the **population risk** or **generalization error**, is $R_D(h) = \Pr_{(x,y) \sim D} [h(x) \neq y]$.

Equivalently,

$$R_D(h) = \mathbb{E}_{(x,y) \sim D} [\mathbf{1}[h(x) \neq y]].$$

This quantity is generally unknown because the distribution D is unknown.

1.2.3 Training Error versus Generalization Error

The central question is therefore

$$\widehat{R}_S(h) \text{ small} \implies R_D(h) \text{ small?}$$

The implication is *not automatically true*.

A sufficiently complex model may memorize the training examples, obtaining nearly zero training error while still making many mistakes on unseen examples. This phenomenon is called **overfitting**.

Learning theory studies the conditions under which the above implication does hold.

1.3 Generalization Gap

The difference between population risk and empirical risk is called the **generalization gap**: $\text{GenGap}(h) = |R_D(h) - \widehat{R}_S(h)|$.

A major objective of learning theory is to establish bounds of the form

$$R_D(h) \leq \widehat{R}_S(h) + \text{complexity term.}$$

The complexity term depends on the richness of the hypothesis class and the number of training examples.

2 Probably Approximately Correct (PAC) Learning

2.0.1 Probably

We want the learning algorithm succeed with probability at least $1 - \delta$ it may not always be with probability one, where confidence parameter $\delta \in (0, 1)$

For example, if $\delta = 0.05$ $1 - \delta = 0.95$. Thus, we require a 95% probability that the desired guarantee holds.

2.0.2 Approximately

We do not require the learned hypothesis to have exactly zero error.

Instead, we specify an accuracy parameter, and allow the desired error is at most ϵ , $\epsilon \in (0, 1)$. For example, $\epsilon = 0.05$ means that we allow a true error of at most 5%.

2.0.3 Correct

The learned hypothesis should satisfy $R_D(h) \leq \epsilon$.

Combining the three components gives the fundamental PAC guarantee: $\Pr_S [R_D(h) \leq \epsilon] \geq 1 - \delta$, where the probability is taken over the random choice of the training sample S . If $\epsilon = 0.05$, $\delta = 0.01$, then

$$\Pr_S [R_D(h) \leq 0.05] \geq 0.99$$

Notice that PAC learning does **not** mean that every future example will be classified correctly.

It means that the *overall probability of making a mistake* on a new example is at most ϵ (5%), with confidence at least $1 - \delta$ (99%).

There is a fundamental trade-off:

smaller ϵ or smaller $\delta \implies$ larger sample size m .

Thus PAC learning is closely related to the question of **sample complexity**.

3 Realizable and Agnostic PAC Learning

3.1 Realizable Setting

In the **realizable** PAC setting, we assume that the true labeling function belongs to the hypothesis class: $f \in \mathcal{H}$. Furthermore, labels are assumed to be generated without noise: $y = f(x)$. Therefore, there exists at least one hypothesis in $\mathcal{H}$ with zero population error.

The goal is to find a hypothesis $h \in \mathcal{H}$ such that $R_D(h) \leq \epsilon$.

3.2 Agnostic Setting

The realizable assumption is often unrealistic.

Real-world data may contain measurement errors, mislabeled examples, ambiguous observations, noise, missing information.

In the **agnostic** setting, we do not assume that the true labeling function belongs to $\mathcal{H}$.

Instead, the goal is to find a hypothesis whose risk is close to the best hypothesis available in the class:

$$R_D(h) \leq \inf_{g \in \mathcal{H}} R_D(g) + \epsilon.$$

Thus,

Realizable: find h with $R_D(h) \leq \epsilon$,

Agnostic: find h nearly as good as the best $g \in \mathcal{H}$.

4 Empirical Risk Minimization

A natural learning strategy is to choose the hypothesis with the smallest training error.

This is called **Empirical Risk Minimization (ERM)**:

$$\hat{h} = \arg \min_{h \in \mathcal{H}} \hat{R}_S(h).$$

However, minimizing training error alone does not guarantee good generalization.

The key issue is therefore:

How complex is the hypothesis class $\mathcal{H}$?

This leads directly to VC dimension.

5 The Hypothesis Class

A learning algorithm generally does not search over every possible function from $\mathcal{X}$ to $\{0, 1\}$.

Instead, it searches within a restricted collection of functions called the **hypothesis class**:

$$\mathcal{H} = \{h : \mathcal{X} \rightarrow \{0, 1\}\}.$$

For example, for linear classification,

$$h_{\mathbf{w}, b}(x) = \begin{cases} 1, & \mathbf{w}^T x + b \geq 0, \\ 0, & \mathbf{w}^T x + b < 0. \end{cases}$$

The corresponding hypothesis class is

$$\mathcal{H} = \left\{ h_{\mathbf{w}, b} : \mathbf{w} \in \mathbb{R}^d, b \in \mathbb{R} \right\}.$$

Different hypothesis classes have different expressive powers.
For example:

Linear classifier	→ relatively restricted,
Decision tree	→ more flexible,
Large neural network	→ highly expressive.

We therefore need a mathematical way to measure the expressive power of $\mathcal{H}$.

6 Vapnik–Chervonenkis (VC) Dimension

It is one of the most important measures of the capacity of a hypothesis class.

The key concept is **shattering**.

6.1 Binary Labelings

Suppose we choose m points: $x_1, x_2, \dots, x_m$. Each point can receive one of two labels: 0 or 1.

Therefore, the total number of possible binary labelings is 2^m .

For example, for $m = 3$ points, there are

$$2^3 = 8$$

possible labelings:

$$000, \quad 001, \quad 010, \quad 011,$$

$$100, \quad 101, \quad 110, \quad 111.$$

6.2 Shattering

Definition 6.1 (Shattering). *A hypothesis class $\mathcal{H}$ is said to **shatter** a set of points*

$$\{x_1, \dots, x_m\}$$

if every possible binary labeling of these points can be realized by some hypothesis in $\mathcal{H}$.

Formally, the set is shattered if $\forall (y_1, \dots, y_m) \in \{0, 1\}^m, \quad \exists h \in \mathcal{H}$ such that $h(x_i) = y_i \quad \forall i$.

Since there are 2^m possible labelings, shattering means that the hypothesis class can realize all 2^m labelings.

7 VC Dimension

Definition 7.1 (VC Dimension). *The **VC dimension** of a hypothesis class $\mathcal{H}$ is the largest number of points that can be shattered by $\mathcal{H}$.*

Formally, $\text{VCdim}(\mathcal{H}) = \max \{m : \exists \{x_1, \dots, x_m\} \text{ shattered by } \mathcal{H}\}$.

If arbitrarily large finite sets can be shattered, then

$$\text{VCdim}(\mathcal{H}) = \infty.$$

i.e. VC dimension measures the capacity of a hypothesis class. A larger VC dimension means that the class can represent a larger variety of labeling patterns.

8 Example: Linear Classifiers

Consider affine linear classifiers in $\mathbb{R}^d$:

$$h_{\mathbf{w},b}(x) = \text{sign}(\mathbf{w}^T x + b).$$

The decision boundary is

$$\mathbf{w}^T x + b = 0.$$

For two-dimensional data, this boundary is a straight line.

A remarkable result is

$$\text{VCdim}(\text{affine linear classifiers in } \mathbb{R}^d) = d + 1.$$

Therefore,

Input dimension d	VC dimension
1	2
2	3
3	4
d	$d + 1$

8.1 Three Points in Two Dimensions

Consider three points in general position in $\mathbb{R}^2$.

There are $2^3 = 8$ possible binary labelings.

For three non-collinear points, a suitable straight line can realize each of these eight labelings.

Therefore, three points can be shattered by affine linear classifiers in $\mathbb{R}^2$.

Hence,

$$\text{VCdim}(\mathcal{H}) \geq 3.$$

On the other hand, four points cannot always be shattered by a straight-line classifier in $\mathbb{R}^2$.

Therefore, $\text{VCdim}(\text{linear classifiers in } \mathbb{R}^2) = 3$.

9 Why VC Dimension Matters

Suppose we have two hypothesis classes: $\text{VCdim}(\mathcal{H}_1) = 5$ and $\text{VCdim}(\mathcal{H}_2) = 1000$. The second hypothesis class has much greater capacity. It can represent substantially more complex decision rules. This is useful because a high-capacity model can fit complicated patterns.

However, it also creates a danger:

$$\text{high capacity} \implies \text{greater potential for overfitting.}$$

A very flexible model may memorize the training data rather than learning the underlying structure of the distribution.

Thus, learning theory seeks a balance between fit to training data and model complexity.

10 Growth Function

VC dimension is connected to another important concept called the **growth function**.

Definition 10.1 (Growth Function). *The growth function of $\mathcal{H}$ is the maximum number of distinct labelings that the hypothesis class can produce on any set of m points.*

It is commonly denoted by $\Pi_{\mathcal{H}}(m)$. Since there are only 2^m possible binary labelings, $\Pi_{\mathcal{H}}(m) \leq 2^m$. If $\mathcal{H}$ shatters some set of m points, then

$$\Pi_{\mathcal{H}}(m) = 2^m.$$

However, once m becomes larger than the VC dimension, the hypothesis class cannot realize all possible labelings.

11 Sauer's Lemma

The connection between VC dimension and the growth function is made precise by Sauer's lemma.

Theorem 11.1 (Sauer's Lemma). *Let $\text{VCdim}(\mathcal{H}) = d$. For $m > d$,*

$$\Pi_{\mathcal{H}}(m) \leq \sum_{i=0}^d \binom{m}{i}.$$

For fixed d , this grows polynomially in m rather than exponentially as 2^m . This is one of the key reasons VC dimension is useful for learning theory.

12 From VC Dimension to Generalization

We can now see the complete chain of ideas.

A hypothesis class $\mathcal{H}$ has a certain expressive capacity.

Its capacity can be measured using VC dimension:

$$\mathcal{H} \longrightarrow \text{VCdim}(\mathcal{H}).$$

VC dimension controls the growth of the number of possible labelings:

$$\text{VCdim}(\mathcal{H}) \longrightarrow \Pi_{\mathcal{H}}(m).$$

The growth function can then be used to establish uniform-convergence and generalization bounds.

A typical VC-style bound has the form

$$R_D(h) \leq \widehat{R}_S(h) + C \sqrt{\frac{d \log(m/d) + \log(1/\delta)}{m}}$$

with probability at least $1 - \delta$, where d is the VC dimension and C denotes a constant depending on the particular version of the bound.

12.1 Interpreting the VC Bound

The bound tells us several important things.

12.1.1 More Training Data

As m increases,

$$\sqrt{\frac{d \log(m/d) + \log(1/\delta)}{m}}$$

generally decreases. Therefore, more data implies better potential generalization.

12.1.2 Higher Model Complexity

As the VC dimension d increases, the complexity term becomes larger. Therefore, higher VC dimension implies more data required for comparable generalization.

12.1.3 Higher Confidence

If we require smaller δ , then $\log \frac{1}{\delta}$ increases. Therefore, demanding higher confidence requires more data.

13 Worked Example: Spam Classification

Suppose a spam classifier is trained on

$$m = 10,000$$

emails.

Let the hypothesis class be the set of linear classifiers based on d features.

The learning algorithm finds a classifier h with

$$\widehat{R}_S(h) = 0.02.$$

Thus, the training error is only 2%. This alone does not tell us that the test error is 2%. We also need to consider the complexity of $\mathcal{H}$. If the hypothesis class has a relatively small VC dimension, a small training error may provide strong evidence of good generalization. If the hypothesis class has extremely large VC dimension, the same training error may provide much weaker evidence. This gives the central learning-theoretic principle: Good training performance is meaningful only relative to model complexity and sample size.

14 Conceptual Questions

1. Why can a classifier have zero training error but non-zero generalization error?
2. What is the difference between empirical risk and population risk?
3. In the PAC guarantee

$$P(R_D(h) \leq \epsilon) \geq 1 - \delta,$$

what do ϵ and δ represent?

4. Why does a smaller ϵ generally require more training examples?
5. What is the difference between the realizable and agnostic PAC settings?
6. What does it mean for a hypothesis class to shatter a set of points?
7. Why can threshold classifiers on $\mathbb{R}$ shatter one point but not two?

8. Why does VC dimension provide a useful measure of model complexity?
9. Why does a larger VC dimension generally require more training data?
10. Explain the relationship

VC dimension, growth function, generalization bound.

“\latex

15 Generalized Linear Models (GLM)

It extends the ordinary linear model by allowing different types of response variables and introducing a link function. 1. Linear Model In an ordinary linear regression model,

$$y = \mathbf{w}^T \mathbf{x} + b + \epsilon$$

or, equivalently,

$$\mathbb{E}[y \mid \mathbf{x}] = \mathbf{w}^T \mathbf{x} + b.$$

The expected output is directly a linear function of the input. 2. Generalized Linear Model: A GLM first computes a linear predictor $\eta = \mathbf{w}^T \mathbf{x} + b$. The expected response is then related to this linear predictor through a link function $g: g(\mu) = \mathbf{w}^T \mathbf{x} + b$ where $\mu = \mathbb{E}[y \mid \mathbf{x}]$. Equivalently, $\mu = g^{-1}(\mathbf{w}^T \mathbf{x} + b)$ where g^{-1} is the inverse link function. 3. Logistic Regression as a GLM

For binary classification,

$$y \in \{0, 1\}.$$

We assume

$$y \sim \text{Bernoulli}(p).$$

The linear predictor is

$$\eta = \mathbf{w}^T \mathbf{x} + b.$$

The logit link function is

$$g(p) = \log \left(\frac{p}{1-p} \right)$$

Therefore,

$$\log \left(\frac{p}{1-p} \right) = \mathbf{w}^T \mathbf{x} + b$$

and the inverse link gives

$$p = \frac{1}{1 + \exp[-(\mathbf{w}^T \mathbf{x} + b)]}$$

which is the sigmoid function. 4. Linear Regression as a Special Case of GLM: For ordinary linear regression,

$$y \sim \mathcal{N}(\mu, \sigma^2)$$

and we use the identity link:

$$g(\mu) = \mu.$$

Therefore,

$$\mu = \mathbf{w}^T \mathbf{x} + b.$$

Thus, Linear Regression is a special case of GLM.

15.1 Why Do We Need Generalized Linear Models?

Linear regression is one of the simplest and most useful statistical models.

In ordinary linear regression, we write

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \epsilon.$$

In matrix notation,

$$Y = X\boldsymbol{\beta} + \epsilon.$$

This model works particularly well when the response variable Y is continuous and approximately Gaussian. However, many machine-learning problems involve response variables that are not naturally modeled by a Gaussian distribution.

For example:

Problem	Response variable	Example
Classification	$Y \in \{0, 1\}$	Spam / not spam
Count prediction	$Y = 0, 1, 2, \dots$	Number of visits
Positive data	$Y > 0$	Insurance claim amount
Proportion	$0 \leq Y \leq 1$	Conversion rate

Using ordinary linear regression for these problems can produce undesirable predictions. For example, a linear regression model for a binary outcome may produce $\hat{y} = -0.4$, $\hat{y} = 1.3$, which cannot be interpreted as probabilities.

This motivates the **Generalized Linear Model (GLM)**.

15.2 The Main Idea of a GLM

The central idea behind a GLM is: Keep the linear predictor, but allow a suitable response distribution and link function.

16 The Exponential Family

16.1 General Form

Let Y follow an exponential family. A distribution belongs to the exponential family if its density or probability mass function can be written in the form

$$p(y \mid \theta, \phi) = \exp \left\{ \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi) \right\}.$$

Here,

- θ is the natural parameter,
- ϕ is the dispersion parameter,
- $b(\theta)$ is the cumulant function,
- $a(\phi)$ and $c(y, \phi)$ are model-specific functions.

Two important results are

$$E[Y] = b'(\theta)$$

and

$$\text{Var}(Y) = b''(\theta)a(\phi).$$

Thus the mean and variance are connected through the exponential-family structure.

16.2 Important GLMs

Some commonly used GLMs are:

Model	Distribution	Link	Typical application
Linear regression	Gaussian	Identity	Continuous prediction
Logistic regression	Bernoulli	Logit	Binary classification
Poisson regression	Poisson	Log	Count prediction
Gamma regression	Gamma	Log / inverse	Positive data

17 Logistic Regression

17.1 Why Logistic Regression?

Suppose

$$Y \in \{0, 1\}.$$

For example,

$$Y = \begin{cases} 1, & \text{spam,} \\ 0, & \text{not spam.} \end{cases}$$

Since Y is binary,

$$Y \sim \text{Bernoulli}(p).$$

We want to predict

$$P(Y = 1 \mid X).$$

Let

$$p = P(Y = 1 \mid X).$$

A probability must satisfy

$$0 \leq p \leq 1.$$

Instead of directly modeling p as a linear function, logistic regression models the **log-odds**.

17.2 Odds and Log-Odds

The odds of an event are

$$\text{Odds} = \frac{p}{1-p}.$$

The log-odds are

$$\text{logit}(p) = \log \frac{p}{1-p}.$$

Logistic regression assumes

$$\log \frac{p}{1-p} = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p.$$

Therefore,

$$\text{logit}(p) = X\beta.$$

Solving for p gives

$$p = \frac{1}{1 + \exp(-X\beta)}.$$

This function is called the **logistic function** or **sigmoid function**.

17.3 Interpretation of Logistic Regression Coefficients

Suppose

$$\log \frac{p}{1-p} = \beta_0 + \beta_1 x.$$

Then increasing x by one unit changes the log-odds by β_1 .

Exponentiating gives

$$e^{\beta_1} = \text{odds ratio for a one-unit increase in } x.$$

17.4 Logistic Regression as a GLM

Logistic regression has:

$$Y \sim \text{Bernoulli}(p)$$

$$\eta = X\beta$$

and

$$\text{logit}(p) = \eta.$$

Thus it satisfies all three GLM components.

18 Maximum Likelihood Estimation in GLMs

The parameters of a GLM are commonly estimated using **maximum likelihood estimation (MLE)**.

Given observations

$$y_1, \dots, y_n,$$

the likelihood is

$$L(\beta) = \prod_{i=1}^n p(y_i \mid x_i, \beta).$$

It is usually easier to maximize the log-likelihood:

$$\ell(\boldsymbol{\beta}) = \log L(\boldsymbol{\beta}).$$

The MLE is

$$\hat{\boldsymbol{\beta}} = \arg \max_{\boldsymbol{\beta}} \ell(\boldsymbol{\beta}).$$

Equivalently, one can minimize the negative log-likelihood:

$$\boxed{\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta}} -\ell(\boldsymbol{\beta}).}$$

For many GLMs there is no simple closed-form solution, so numerical optimization methods are used.

Common methods include:

- Newton–Raphson,
- Fisher scoring,
- Iteratively Reweighted Least Squares (IRLS).

19 Regression with Correlated Data

19.1 Why Correlation Matters

Ordinary regression often begins with

$$Y = X\boldsymbol{\beta} + \epsilon$$

and assumes

$$E[\epsilon] = 0$$

and

$$\text{Var}(\epsilon) = \sigma^2 I.$$

The second assumption means that the errors are uncorrelated and have equal variance. However, many real datasets contain observations that are naturally related.

Examples include:

- repeated measurements of the same patient,
- multiple observations from the same student,
- measurements over time,
- observations from the same geographical region,
- customers belonging to the same company,
- multiple measurements from the same machine.

In such situations,

$$\text{Cov}(\epsilon_i, \epsilon_j) \neq 0$$

may hold.

We therefore write

$$\text{Var}(\epsilon) = \Sigma$$

rather than

$$\sigma^2 I.$$

19.2 A Simple Example of Correlated Data

Suppose a doctor measures a patient's blood pressure at five different times.

The observations might be

$$Y_1, Y_2, Y_3, Y_4, Y_5.$$

These measurements are not independent.

If the patient's blood pressure is unusually high because of an individual characteristic, several measurements may be high.

Thus,

$$\text{Cov}(Y_i, Y_j) > 0$$

for measurements belonging to the same patient.

The same idea appears in longitudinal machine-learning datasets.

19.3 What Goes Wrong with Ordinary OLS?

The ordinary least-squares estimator is

$$\hat{\beta}_{OLS} = (X^T X)^{-1} X^T Y.$$

Under appropriate exogeneity conditions, OLS can remain unbiased even when errors are correlated.

However, two important problems can arise:

1. OLS is generally no longer the most efficient linear unbiased estimator.
2. The usual IID standard-error formula can be incorrect.

Therefore, statistical inference based on naive OLS standard errors may be misleading.

The problem is not simply that “OLS always gives the wrong coefficients.” The more precise issue is that the covariance structure of the errors has not been correctly accounted for.

20 Generalized Least Squares

20.1 The GLS Model

Consider

$$Y = X\beta + \epsilon$$

with

$$E[\epsilon] = 0$$

and

$$\text{Var}(\epsilon) = \Sigma.$$

The matrix Σ describes the variances and covariances of the errors.

If Σ is known and positive definite, we can account for the correlation using **Generalized Least Squares (GLS)**.

20.2 GLS Objective Function

GLS minimizes the covariance-weighted squared error:

$$Q(\beta) = (Y - X\beta)^T \Sigma^{-1} (Y - X\beta).$$

Notice that ordinary least squares minimizes

$$(Y - X\beta)^T (Y - X\beta),$$

while GLS uses

$$\Sigma^{-1}.$$

Thus observations are weighted according to the covariance structure.

20.3 Derivation of the GLS Estimator

We minimize

$$Q(\beta) = (Y - X\beta)^T \Sigma^{-1} (Y - X\beta).$$

Taking the derivative with respect to β gives

$$\frac{\partial Q}{\partial \beta} = -2X^T \Sigma^{-1} (Y - X\beta).$$

Setting the derivative equal to zero,

$$X^T \Sigma^{-1} (Y - X\beta) = 0.$$

Therefore,

$$X^T \Sigma^{-1} Y = X^T \Sigma^{-1} X \beta.$$

Hence,

$$\hat{\beta}_{GLS} = (X^T \Sigma^{-1} X)^{-1} X^T \Sigma^{-1} Y.$$

This is one of the most important results for regression with correlated errors.

20.4 Covariance of the GLS Estimator

Under the model assumptions,

$$\text{Var}(\hat{\beta}_{GLS}) = (X^T \Sigma^{-1} X)^{-1}.$$

Compare this with ordinary least squares:

$$\text{Var}(\hat{\beta}_{OLS}) = \sigma^2 (X^T X)^{-1}$$

under the IID homoscedastic assumption.

20.5 GLS Reduces to OLS

Suppose

$$\Sigma = \sigma^2 I.$$

Then

$$\Sigma^{-1} = \frac{1}{\sigma^2} I.$$

Substituting into the GLS estimator,

$$\hat{\beta}_{GLS} = (X^T X)^{-1} X^T Y.$$

Therefore,

OLS is a special case of GLS.

This is an important conceptual connection.

21 Overview of Regularization

21.1 Why Do We Need Regularization?

Suppose we have a model with many parameters.

For example,

$$Y = X\beta + \epsilon$$

with

$$p \gg n.$$

There may be many possible parameter vectors that fit the training data extremely well. A highly flexible model may have

$$\text{training error} \approx 0$$

but

$$\text{test error} \gg 0.$$

This is the problem of **overfitting**.

Regularization provides a systematic way of controlling model complexity.

21.2 The General Regularized Objective

Instead of minimizing only empirical loss,

$$\min_{\boldsymbol{\beta}} \widehat{L}(\boldsymbol{\beta}),$$

we add a penalty:

$$\min_{\boldsymbol{\beta}} \left[\widehat{L}(\boldsymbol{\beta}) + \lambda R(\boldsymbol{\beta}) \right].$$

Here,

- $\widehat{L}(\boldsymbol{\beta})$ is the training loss,
- $R(\boldsymbol{\beta})$ is the regularization penalty,
- $\lambda \geq 0$ controls the strength of regularization.

21.3 Effect of λ

The parameter λ controls the strength of the penalty.

If

$$\lambda = 0,$$

Very large λ can make the model too simple and cause underfitting.

Thus there is typically a useful intermediate value of λ .

22 Ridge Regression

22.1 L2 Regularization

Ridge regression uses an L2 penalty:

$$R(\boldsymbol{\beta}) = \|\boldsymbol{\beta}\|_2^2 = \sum_{j=1}^p \beta_j^2.$$

The ridge objective is

$$\min_{\boldsymbol{\beta}} \left[\|Y - X\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_2^2 \right].$$

22.2 Ridge Solution

Taking the derivative and setting it to zero gives

$$-2X^T(Y - X\boldsymbol{\beta}) + 2\lambda\boldsymbol{\beta} = 0.$$

Therefore,

$$X^T X \boldsymbol{\beta} + \lambda \boldsymbol{\beta} = X^T Y.$$

Hence,

$$\hat{\boldsymbol{\beta}}_{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T Y.$$

This is the closed-form ridge estimator.

22.3 Why Does Ridge Help?

When predictors are highly correlated,

$$X^T X$$

may be ill-conditioned or nearly singular.

Ridge replaces it with

$$X^T X + \lambda I.$$

This improves numerical stability and shrinks large coefficients.

Thus ridge is especially useful for:

- multicollinearity,
- high-dimensional predictors,
- unstable coefficient estimates,
- reducing variance.

Ridge usually makes coefficients smaller but does not make them exactly zero.

23 Lasso Regression

23.1 L1 Regularization

Lasso uses an L1 penalty:

$$R(\boldsymbol{\beta}) = \|\boldsymbol{\beta}\|_1 = \sum_{j=1}^p |\beta_j|.$$

The lasso objective is

$$\min_{\boldsymbol{\beta}} [\|Y - X\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_1].$$

23.2 Why Does Lasso Produce Sparse Models?

The absolute-value penalty has a sharp corner at

$$\beta_j = 0.$$

As a result, optimization frequently places coefficients exactly at zero.

Therefore, Lasso performs both estimation and variable selection.

23.3 Soft Thresholding

The sparsity property can be understood from the one-dimensional problem

$$\min_{\beta} \left[\frac{1}{2}(z - \beta)^2 + \lambda |\beta| \right].$$

Its solution is

$$\hat{\beta} = \text{sign}(z) (|z| - \lambda)_+,$$

where

$$(a)_+ = \max(a, 0).$$

Thus, if

$$|z| \leq \lambda,$$

then

$$\hat{\beta} = 0.$$

This is the mathematical reason Lasso creates sparsity.

24 Elastic Net

Ridge and Lasso have different strengths.

Method	Penalty	Main property
Ridge	L2	Stable shrinkage
Lasso	L1	Sparsity / variable selection
Elastic Net	L1 + L2	Sparsity + stability

Elastic Net combines both penalties:

$$\min_{\beta} \left[\hat{L}(\beta) + \lambda \left(\alpha \|\beta\|_1 + \frac{1-\alpha}{2} \|\beta\|_2^2 \right) \right].$$

where

$$0 \leq \alpha \leq 1.$$

Special cases are:

$$\alpha = 1 \quad \Rightarrow \quad \text{Lasso},$$

and

$$\alpha = 0 \quad \Rightarrow \quad \text{Ridge}.$$

Elastic Net is particularly useful when there are groups of highly correlated predictors.

25 Geometric Interpretation of Regularization

25.1 Ridge Geometry

Ridge constrains the coefficients using $\|\beta\|_2^2 \leq t$. In two dimensions this is a circle. The optimum usually touches the circle away from the coordinate axes. Therefore coefficients are generally nonzero.

25.2 Lasso Geometry

Lasso uses $\|\beta\|_1 \leq t$. In two dimensions this is a diamond. The diamond has corners on the coordinate axes. Because the optimum frequently occurs at a corner, $\beta_1 = 0$, $\beta_2 = 0$. can occur naturally.

26 Bias–Variance Trade-off

Regularization changes the bias and variance of an estimator. For squared prediction error at a fixed input x ,

$$E[(\hat{f}(x) - f(x))^2] = \text{Bias}[\hat{f}(x)]^2 + \text{Var}[\hat{f}(x)] + \sigma^2.$$

Here: Bias measures systematic error, Variance measures sensitivity to the training sample σ^2 represents irreducible noise.

Increasing regularization generally causes high bias and lower variance. The goal is not to minimize training error.

The goal is to choose a model that gives good performance on unseen data. Therefore, Regularization trades a small increase in bias for a potentially large reduction in variance.

27 Regularization for Logistic Regression

Regularization is not limited to linear regression. For logistic regression, the objective can be written as

$$\min_{\beta} -\ell(\beta) + \lambda \|\beta\|_2^2.$$

This gives **ridge logistic regression**.

Alternatively,

$$\min_{\beta} -\ell(\beta) + \lambda \|\beta\|_1$$

gives **lasso logistic regression**. Regularization is particularly useful when: the number of features is large, $p \gg n$, predictors are highly correlated, the data are nearly or completely separable.

28 Conceptual Questions

1. Why can ordinary linear regression produce invalid predictions for binary outcomes?
2. What are the three components of a GLM?
3. What is the purpose of the link function?
4. Why is the logit link suitable for binary classification?
5. Interpret e^{β_j} in logistic regression.
6. Why is the log link useful in Poisson regression?
7. What is the difference between a probability model and a link function?
8. What changes when

$$\text{Var}(\epsilon) = \sigma^2 I$$

is replaced by

$$\text{Var}(\epsilon) = \Sigma?$$

9. Why can OLS remain unbiased under correlated errors but still be inefficient?
10. Derive the GLS estimator.

11. Show that OLS is a special case of GLS.
12. What is the difference between fixed and random effects?
13. What does ICC measure?
14. Why does L2 regularization shrink coefficients?
15. Why can L1 regularization produce exact zeros?
16. Explain the geometric difference between Ridge and Lasso.
17. Explain regularization using the bias–variance trade-off.
18. Explain the Bayesian interpretation of Ridge and Lasso.
19. Why should λ be selected using validation data rather than the final test set?
20. Why is ordinary random cross-validation potentially problematic for correlated or grouped data?